Total number of printed pages–3

4 SEM MCA (NCBCS) SE 3

2019

(June)

COMPUTER APPLICATION

Paper : 403

(Software Engineering)

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

*Answer **any five** questions.*

1. *(a)* Explain Waterfall model with advantages and disadvantages. 8

   (b) Explain the Software requirement specification (SRS) document. 7

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2. (a) Discuss the development of basic Cocomo model and state the principles of effort estimation of a software system. 8

(b) What are the characteristics of a modular system ? Define and compare Coupling and Cohesion. 7

3. (a) Define Software prototyping. Describe briefly the different types of prototyping. What are the advantages of developing a prototype of a system ? 8

(b) Define Re-engineering. Explain the Re-engineering process model. 7

4. (a) What are the different levels of testing ? Explain *any one* with suitable example. 8

(b) What is the ISO 9000 certification ? What are the types of ISO 9000 quality standards ? Explain the need for obtaining ISO 9000 certification. 1+2+4=7

5. (a) Explain briefly Code Walkthrough and Code Inspection. 4+4=8

(b) Define Software Reliability. Explain Software-reliability Specification. What is the different classification o software failure ?

6. (a) Explain Belady and Lehman model fo Software maintenance.

(b) Explain Evolutionary softwa development model in details.

(c) State *at least six* advantages of Obje oriented design technique.

(d) Differentiate between Softwa products and programs.

7. Write short notes on : ***(any three)*** 3×5=

(a) Spiral model

(b) Complete Cocomo model

(c) White Box Testing

(d) Software faults and failure.

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Total number of printed pages–4

4 SEM MCA (NCBCS) SE 3

2018

(June)

COMPUTER APPLICATION

Paper : 403

(Software Engineering)

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

*Answer **any five** questions.*

1. *(a)* Give the stages of development of waterfall model in details. Give a comparison of this model with spiral model. 5+3=8

   (b) What is Modularization ? Explain the principle involved in Software coupling and Software cohesion. 7

Contd.

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2. (a) What are the various tools used in Software analysis and design ? Give objectives of each of them. 3

(b) Differentiate between flowchart and DFD. 5

(c) Explain giving a neat diagram the RAD model with its advantages and disadvantages. 7

3. (a) Distinguish between Top-down approach and Bottom-up approach. 5

(b) Distinguish between Software verification and validation. 5

(c) Distinguish between Conceptual design and Technical design of software. 5

4. (a) Explain the objectives of Software Requirement Specification (SRS).

List the characteristics of good SRS. 3+3=6

(b) Explain all the phases involved in the development of COCOMO model. 5

(c) State in your words, why Software needs maintenance. 4

5. (a) What are the different levels of Reverse Engineering ? Explain in detail all the tasks in Reverse Engineering. 8

(b) Mention the typical risk factors that threaten a project in different ways.

Explain the various risk management activities in Software Engineering. 7

6. (a) Explain various object-oriented concepts used in Software Engineering

(b) Explain various Reliability Matrices.

(c) What do you mean by System Testing What are its different types ? Explain

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7. Write short notes on : ***(any three)*** 3×5=15

(a) Configuration Management

(b) Evolutionary model

(c) Prototype model

(d) Market assessment of Software.

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Total No. of printed pages = 3

4 SEM MCA SE 3 (NCBCS)

2016

(June)

COMPUTER APPLICATION

Paper : MCA-403

(Software Engineering)

Full Marks – 75

Pass Marks – 34

Time – Three hours

The figures in the margin indicate full marks for the questions.

Answer *all* the questions.

1. (a) What are the important activities that are carried out during the feasibility study phase ? 3

   (b) Distinguish the spiral model with the prototype model with their advantages and disadvantages. 7

   (c) Explain the advantages and disadvantages of top-down testing. 3

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(d) State two requisites that a project team should have. 2

2. (a) Compare the relative advantages of waterfall model, prototype model and spiral model. Which model would you prefer in case when a customer frequently changes his requirements ? Give reasons for your answer. 5+3=8

(b) What are the major sources of risk ? Explain the types and categories of risks that a software project may suffer from. 7

3. (a) Discuss the different types of coupling and cohesion measures of modules. 8

(b) Distinguish between structural design and object oriented design. 3

(c) List the characteristics of a good SRS. 4

4. (a) Discuss the relationship between software configuration management and software maintenance. 7

(b) Distinguish between : 4+4=

(i) Software verification and validation.

(ii) Software faults and failures.

5. (a) What is Re-engineering ? Explain the engineering process model.

(b) Write short notes on : 5+5=

(i) Boehm's model for software maintenan

(ii) Partitioning and abstraction.

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Total number of printed pages–3

4 SEM MCA SENGG 3

2017

(June)

COMPUTER APPLICATION

Paper : 403

(Software Engineering)

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

*Answer **any five** questions.*

1. *(a)* Explain the Classical Waterfall model with advantage and disadvantage. 6

   (b) Mention the differences between Prototyping model and Evolutionary model. 4

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(c) What are the stages of SDLC ? Explain in details. 5

2. (a) Explain the Intermediate COCOMO and BASIC COCOMO in details. 6

(b) List *four* reasons why it is difficult to improve software process. 4

(c) Explain the Risk Containment Phase and Risk Leverage Phase. 5

3. (a) What do you mean by Modularization ? Mention its advantages. 5

(b) What are the stages involved in Software Project Management Plan ? 4

(c) What are the categories of Software Maintenance ? Mention the need for Maintenance. 6

4. (a) What are the types of Integration Testing Approach ? Explain each. 6

(b) How do Hardware reliability differ from Software Reliability ? Explain. 4

(c) Explain the code inspection techniques for code review. 5

5. (a) What is the ISO 9000 Certification What are the types of ISO 9000 qualit standards ? Explain the need fo obtaining ISO 9000 certification. 2+3+4=

(b) What are the software Debuggin approaches ? Explain. 6

6. (a) Explain the roles of a Systems Analyst 4

(b) What do you mean by Unit Testing anc Black Box Testing ? 4

(c) What are Case tools ? Explain the building blocks of Case tools. 7

7. Write short notes on : ***(any three)*** 5×3=15

I. Boehm's model for software maintenance

II. Line of codes

III. White-box testing

IV. Spiral model

V. Partitioning and Abstraction.

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